



Predictive Modelling with the Open University Learning Analytics Dataset (OULAD): A Systematic Literature Review

Lingxi Jin¹ , Yao Wang², Huiying Song³, and Hyo-Jeong So¹  

¹ Department of Educational Technology, Ewha Womans University, Seoul, South Korea
{jlinlingxi,hyojeongso}@ewha.ac.kr

² National Institute of Education, Nanyang Technological University, Singapore, Singapore
nie23.wy3836@e.ntu.edu.sg

³ Department of Computer Science, Ewha Womans University, Seoul, South Korea
shy114@ewha.ac.kr

Abstract. Higher education has experienced an unparalleled digital transformation, driven by the widespread adoption of online learning with massive users, which has risen to an explosive growth in the generation and analysis of student-related data. Within this transformation, predictive modeling has emerged as a useful tool for predicting critical indicators in the learning process, encompassing students' academic performance, class retention, and dropout rates. With this backdrop, this study aims to conduct a systematic review of recent publications focused on predictive modeling, with a specific emphasis on the Open University Learning Analytics Datasets (OULAD). Following the PRISMA process, we identified 17 research articles published from 2017 to 2024, concentrating on OULAD in higher education. For our analysis, we categorized the purpose of predictive modeling into three types: (a) predicting students' performance, (b) identifying at-risk students, and (c) predicting student engagement. The central focus lies on the identification of algorithms predominantly employed in these studies, including machine learning, deep learning, and statistical models. By investigating the methodologies and algorithms employed, this review informs researchers in learning analytics and educational data mining of the potential opportunities and challenges associated with predictive modeling using OULAD in higher education.

Keywords: Predictive modelling · Open University Learning Analytics Dataset (OULAD) · Educational data mining (EDM)

1 Introduction

With the widespread adoption of Massive Open Online Courses (MOOCs), the scale of student data collection has reached unprecedented levels. Educational institutions now have access to vast amounts of data on students' learning patterns, preferences, and progress. This wealth of data enables educators to personalize instruction, ultimately enhancing student engagement and success. This digital revolution has not only transformed the delivery methods of instruction but has also given rise to the field dedicated to

examining learner data. Learning analytics [4], academic analytics and educational data mining [7] are three prominent fields in the domain of leveraging learner-generated data and analytical models to uncover valuable insights into learning processes. In particular, learning analytics focuses on utilizing these data to unpack learning processes and social connections while making predictions and recommendations about learning outcomes [18].

Within this context, the Open University Learning Analytics Dataset (OULAD) which is publicly accessible, becomes a valuable resource for predictive modeling. OULAD includes a wealth of learning-related data, such as study duration, access frequency, and course grades, offering researchers valuable insights into student learning behaviors [13]. The data of OULAD is utilized to model training that is authentic and reliable, supporting the development of future virtual learning environments and enabling other educational institutions to use the data to innovate and develop their instructional methods. However, several challenges persist in the practical implementation of predictive modeling, including the quality and integrity of data, the efficacy of feature selection, as well as the complexity and interpretability of models [13].

With this backdrop, this paper aims to provide a systematic literature review of predictive modeling research using OULAD. By investigating the methods and algorithms employed, this review seeks to inform researchers in learning analytics and educational data mining of the potential opportunities and challenges associated with predictive modeling using OULAD in higher education. The research questions that guided this study are as follows: **RQ1:** What are the research trends of the predictive modeling using OULAD? **RQ2:** What are the research themes in predictive modeling using OULAD? **RQ3:** What prediction models are used in OULAD in the literature?

2 Methods

2.1 Data Sources and Search Strategy

This systematic review followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) 2020 guidelines to ensure the rigor in reporting [16]. A wide range of databases was used as our primary source, including IEEE Xplore, Scopus, ScienceDirect, and Web of Science. ERIC and ProQuest were included as additional sources to expand the literature in the educational context relevant to our study. We selected these databases since most publication channels led by the Society of Learning Analytics Research (SoLAR), the International Educational Data Mining Society (IEDMS), and the International Artificial Intelligence in Education Society (IAIED) are available in these databases. The literature search was conducted in Dec 2023 using two sets of search terms as shown in Table 1. The first set of search terms consists of different variations of the term OULAD. The second set includes terms related to predictive modelling. In total, 3,062 papers that contain the two sets of search terms within the title and abstract were identified in the initial search process.

Table 1. Search terms used in each category.

Categories	Search terms in the category
Open University Learning Analytics Dataset	OULAD, OUL*, OULA* dataset, Open University, Learning analytics
Predictive modelling	Prediction model, Predictive analytics, Educational Data Mining

2.2 Inclusion and Exclusion Criteria

The overview of the search protocol is summarized in Fig. 1. This protocol adhered to the PRISMA process [16] and was reconstructed utilizing the template provided by Page et al. [16]. To identify suitable studies for this review, we employed the following inclusion and exclusion criteria. First, we limited the publication period from the beginning of public access to the OULAD in 2017 to January 2024 to ensure the relevance of the literature. Next, we limited the search to peer-reviewed journal articles, case studies, and conference proceedings for quality assurance. Duplicate copies of the studies were removed. Based on this criterion, books and review articles were also excluded from the search. The search was further limited to papers published in English to match the authors’ language proficiency, reducing the search to 1,021 papers. Next, the titles and abstracts were scanned to ensure that the chosen articles were relevant to the main goal of this research. By this criterion, we excluded studies such as the use of non-OULAD for predictions (e.g., the use of the International University of La Rioja educational dataset to predict learner performance by [10] and non-education contexts [22]. This reduced the number of shortlisted articles to 29. The full text of each study was reviewed and 12 articles that did not use OULAD for predictive modelling were further removed. This PRISMA process led to the identification 17 articles eligible for the final analysis.

2.3 Open University Learning Analytics Dataset

Open University Learning Analytics Dataset includes 22 modules with 32,593 Open University registered students. It mainly contains three types of information: demographic data, student performance data, and learning behavior data [3]. First, the demographic data covers basic information about students, including age, gender region, and previous education background. These data are crucial for understanding students’ learning backgrounds and characteristics. Second, student performance data mainly reflects students’ learning achievements and results during their studies at OU, including their evaluation results. These data help researchers understand students’ learning progress and mastery levels. Third, learning behavior data documents students’ online learning activities, such as daily clicks, which can reveal students’ learning habits and patterns.

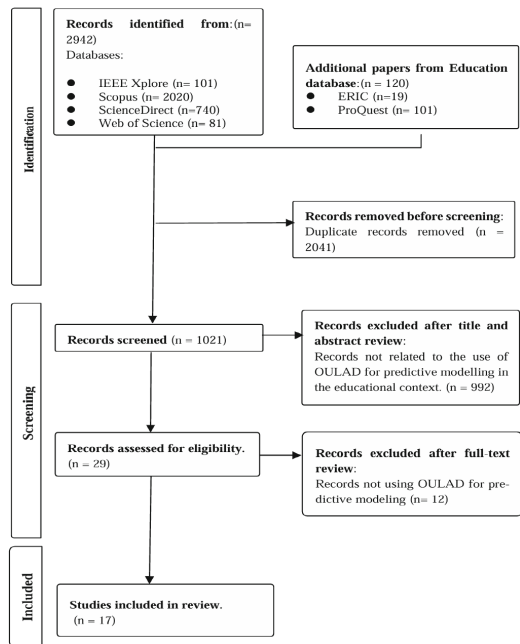


Fig. 1. PRISMA Flowchart for Presented Systematic Review (Framework adapted and modified from Page et al.).

3 Findings

3.1 Bibliometric Analysis

The results reveal a trend in the publication timeline, with all 17 papers collected being published after 2021. This indicates a surge in interest in using the OULAD for predictive modelling in recent years. The year 2022 stood out as the highest number of papers to the field ($n = 7$). Regarding the type of publication, 17 articles were evenly distributed, with 9 published in academic journals and 8 published in conference proceedings.

3.2 Predictive Models

Three overarching themes emerged as the main focus of predictive models: (a) student performance, (b) students at risk and (c) student engagement, as illustrated in Table 2. In this section, we describe the diverse aspects of predictive modelling based on the OULAD employed in the research and the results obtained.

Student Performance Prediction. The first theme, student performance, focuses on the classification of final grades and the prediction of students' pass or fail outcomes in particular courses. The studies employed various models and techniques including the BART model [8], RNN + LSTM [12], Logistic Regression [14, 15], K-Nearest neighbour [15, 17], random forest [8, 26], Gradient-Boosted Tree [15], 1D-CNN [15],

Table 2. Aspect of the predictive modelling based on OULAD.

Theme	Code	Authors	Years	Dataset Features	Algorithms	Best Accuracy
Student Performance	C1. Student Academic Performance	Almahdi et al. [2]	2023	Engagement, Demographic, Performance	MLP, ANN, NB, SVM	MLP (98.94%)
		Drousiotis et al. [8]	2021	Engagement, Performance	BART, RF, DT	BART (92%)
		Al-Tameemi et al. [3]	2021	Interaction	DL	-
		Kukkar et al. [12]	2024	Behavior, Engagement	RNN, LSTM, ML	RNN + LSTM + RF (96.78%)
		Liu et al. [15]	2022	Behavior, Performance	LSTM, 1D-CNN, RF, GBT, K-NN, LR	LSTM (90.25%)
		Lakshmi et al. [14]	2023	Relationship, Performance	GCN, SVM, LR, FFN	PRISM+GCN (almost 82%)
		Qiu et al. [17]	2022	Engagement, Demographic, Performance and Emotion	SVC(R), SVC(L), Naive Bayes, KNN(U), KNN(D), and Softmax	BCEP Model (97.4%)
		Wang et al. [26]	2022	Demographic, Performance, Behavior	Light GBM, XGBoost, GBDT (Gradient Boosting DT, RF, AdaBoost)	-
	C2. Early Prediction	Tonghui. [24]	2023	Demographic, Performance	ML	-
	C3. Risk Students Performance	Souai. [20]	2022	Sequential trajectories, Behavior	BLSTM	96.90%
At-risk student	C4. MOOC Performance	Hao et al. [11]	2022	Performance Behavior	LSTM, 1D-CNN	LSTM (90.25%)
	C5. Student Dropouts	Adnan et al. [1]	2021	Engagement, Performance, Behavior	RF, SVM, ET, AdaBoost, Gradient Boost	RF (91%)
		Shafiq et al. [21]	2022	Demographic, Behavior	LSTM, RTV-SVM, RF, ANN, J48, DT, XGBoost, Naive Bayes	LSTM (97.25%)
		Ali et al. [5]	2021	Engagement, Performance	Ada Boost, RF, SVM, LR, KNN	-
	C6. Learning Trajectories	Gupta et al. [9]	2022	Engagement, Demographic, Behavior	HMM, LR	HMM (94.83%)
	C7. Predict Student Success	Waheed et al. [27]	2023	Engagement, Demographic, Performance	LSTM, GBT, SVM, ANN, LR, DT, KNN	LSTM (89.77%)
	C8. Student Participation	Raj et al. [18]	2022	Behavior, Performance, Interaction	RM, KNN, DT, LR	RF (94.12%)

GCN [14], and Light GBM [26] to predict student academic performance based on diverse datasets such as behavioral data, demographic data, and interaction logs. For instance, Almahdi and Sharef [2] proposed an MLP network model, while Drousiotis et al. [8] employed the BART model, both of which effectively processed complex datasets including behavioural and demographic data to predict student performance. These models successfully predicted whether students passed or failed courses. Additionally, Kukkar et al. [12] demonstrated the superiority of LSTM algorithms in predicting student academic performance by integrating RNN + LSTM feature extraction techniques and various machine learning models.

At-risk Student Prediction. The second theme focuses on identifying at-risk students, understanding the factors that affect student performance, and early prediction of dropouts and withdrawals during or after a course. Regarding student dropouts, Adnan et al. [1] examined the challenges faced by online learners, emphasizing high dropout rates and low engagement levels. Their proposed predictive model, leveraging machine learning and deep learning algorithms, identified at-risk students, thereby enabling effective intervention strategies. Similarly, Shafiq et al. [21] introduced a machine learning-based predictive model focused on identifying students prone to attrition in virtual

learning environments. Additionally, Ali et al. [5] examined the issue of dropout rates in MOOCs by employing sequential classification approaches and comparing various classification models to predict the likelihood of student dropout rates.

Student Engagement. The final theme, student engagement, indicates the degree of interaction and active participation of students during the learning process. For instance, Raj et al. [18] predicted students' engagement in a course by analyzing data on their behaviours in a virtual learning environment (VLE). The researcher utilized various features, including student activities, interactions, and scores in the VLE, to construct a robust model that accurately predicts student engagement. Through experimentation, it was determined that the Random Forest classifier yielded optimal results within the current setup. Leveraging the Random Forest classification algorithm resulted in an impressive accuracy rate of 95%, precision of 95%, and correlation coefficient of 98%.

4 Discussion and Conclusion

This paper systematically reviews recent research in predictive modelling using the OULAD to identify current trends and advances in the field. The findings indicate that most of the existing publications use predictive modelling to assess student performance and predict those students who are at risk of failing or dropping out of school. At the same time, less attention has been paid to student engagement. In terms of the techniques used for prediction, machine learning models including decision trees, random forest, support vector machines (SVM), logistic regression, gradient boosting algorithms, and k-nearest neighbours (KNN) have been used with significant frequency. Statistical models such as linear regression and generalized linear models have also been used in some studies to explore the relationship between predictor variables and outcomes of interest. Deep learning models including neural networks convolutional neural networks (CNNs), recurrent neural networks (RNNs), and auto encoders have gained popularity in recent years for their ability to handle large-scale data and extract meaningful patterns from complex data. The results of these studies vary depending on the chosen model, the characteristics of the dataset, and the specific research question. Generally, machine learning and deep learning models demonstrate good predictive performance in most cases, especially when dealing with complex and non-linear relationships in the data. However, it is essential to consider the limitations of each model and to evaluate their performance using appropriate evaluation methods.

Based on the key findings, we present some areas for future research directions. First, researcher need to consider the potential difference of predictive power and accuracy due to the limited availability of training data and the lack of interpretability of the model. Future research should not only focus on the accuracy and performance of the model, but also examine the interpretability and explain ability of the model. This will help increase trust in the model's predictions and make it easier for educators to understand and accept the model's results. At the same time, efforts should be made to improve the quality and quantity of training data to reduce the performance differences of models in different educational scenarios. Second, it is necessary to strengthen the integration of educational background and theoretical knowledge with machine learning models to

improve the model's ability to understand and solve educational problems. As Baker and Yacef [6] aptly pointed out, understanding students' educational backgrounds, applying relevant theories, and considering various phenomena in educational settings are crucial for effective performance prediction. Therefore, future research should comprehensively consider these factors to enhance the accuracy and practicality of predictive models. Third, it is important to note that the literature surveyed lacks empirical testing of predictive models in real instructional contexts and other virtual learning environments. Hence, future research should address this gap by conducting empirical studies to evaluate predictive models' effectiveness in diverse educational settings.

References

1. Adnan, M., et al.: Predicting at-risk students at different percentages of course length for early intervention using machine learning models. *IEEE Access*. **9**, 7519–7539 (2021)
2. Almahdi, A.A., Sharef, B.T.: Deep learning based an optimized predictive academic performance approach. In: 2023 International Conference on IT Innovation and Knowledge Discovery, pp. 1–6. IEEE (2023)
3. Al-Tameemi, G., et al.: A deep neural network-based prediction model for students' academic performance. In: 2021 14th International Conference on Developments in eSystems Engineering, pp. 364–369. IEEE (2021)
4. Avella, J.T., Kebritchi, M., Nunn, S.G., Kanai, T.: Learning analytics methods, benefits, and challenges in higher education: a systematic literature review. *Online Learn.* **20**(2), 13–29 (2016)
5. Ali, H.A., Mohamed, C., Abdelhamid, B., El Alami, T.: Prediction MOOC's for student by using machine learning methods. In: 2021 XI International Conference on Virtual Campus, pp. 1–3. IEEE (2021)
6. Barber, R., Sharkey, M.: Course correction: using analytics to predict course success. In: Proceedings of the 2nd International Conference on Learning Analytics and Knowledge, pp. 259–262. ACM (2012)
7. Campbell, J.P., DeBlois, P.B., Oblinger, D.G.: Academic analytics: a new tool for a new era. *EDUCAUSE Rev.* **42**(4), 40 (2007)
8. Drousiotis, E., Shi, L., Maskell, S.: Early predictor for student success based on behavioural and demographical indicators. In: Cristea, A.I., Troussas, C. (eds.) ITS 2021, LNCS, vol. 12677, pp. 161–172. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-80421-3_19
9. Gupta, A., Garg, D., Kumar, P.: Mining sequential learning trajectories with hidden Markov models for early prediction of at-risk students in e-learning environments. *IEEE Trans. Learn. Technol.* **15**(6), 783–797 (2022)
10. Hidalgo, A.C., Ger, P.M., Valentin, L.D.L.F.: Using Meta-Learning to predict student performance in virtual learning environments. *Appl. Intell.* **52**(3), 1–14 (2022)
11. Hao, J., Gan, J., Zhu, L.: MOOC performance prediction and personal performance improvement via Bayesian network. *Educ. Inf. Technol.* **27**(5), 7303–7326 (2022)
12. Kukkar, A., Mohana, R., Sharma, A., Nayyar, A.: A novel methodology using RNN + LSTM + ML for predicting student's academic performance. *Educ. Inf. Technol.* (2024). <https://doi.org/10.1007/s10639-023-12394-0>
13. Kuzilek, J., Hlosta, M., Zdrahal, Z.: Open university learning analytics dataset. *Sci. Data* **4**(1), 1–8 (2017)
14. Lakshmi, P. R., Geetha, A. V., Priyanka, D., Mala, T.: PRISM: predicting student performance using integrated similarity modeling with graph convolutional networks. In: 2023 12th International Conference on Advanced Computing, pp. 1–7. IEEE (2023)

15. Liu, Y., Fan, S., Xu, S., Sajjanhar, A., Yeom, S., Wei, Y.: Predicting student performance using clickstream data and machine learning. *Educ. Sci.* **13**(1), 17 (2022)
16. Page, M.J., et al.: The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Int. J. Surg.*, 88 (2021)
17. Qiu, F., et al.: Predicting students' performance in e-learning using learning process and behaviour data. *Sci. Rep.* **12**(1), 453 (2022)
18. Raj, N.S., Renumol, V.G.: Early prediction of student engagement in virtual learning environments using machine learning techniques. *E-Learn. Digital Media* **19**(6), 537–554 (2022). <https://doi.org/10.1177/20427530221108027>
19. Ranjeeth, S., Latchoumi, T.P., Victor Paul, P.: A survey on predictive models of learning analytics. *Procedia Comput. Sci.* **167**, 37–46 (2020). <https://doi.org/10.1016/j.procs.2020.03.180>
20. Souai, W., et al.: Predicting at-risk students using the deep learning BLSTM approach. In: 2022 2nd International Conference of Smart Systems and Emerging Technologies, pp. 32–37. IEEE (2022)
21. Shafiq, D.A., Marjani, M., Habeeb, R.A.A., Asirvatham, D.: A conceptual predictive analytics model for the identification of at-risk students in VLE using machine learning techniques. In: 2022 14th International Conference on Mathematics, Actuarial Science, Computer Science and Statistics, pp. 1–8. IEEE (2022)
22. Spadon, G., et al.: Pay attention to evolution: time series forecasting with deep graph-evolution learning. *IEEE Trans. Pattern Anal. Mach. Intell.* **44**(9), 5368–5384 (2021)
23. Tonghui, X.: Using data mining models to predict students' academic performance before the online course start. *J. Educ. Online* **20**(1), 222–235 (2023). <https://doi.org/10.9743/jeo.2023.20.1.15>
24. Torres Martín, C., Acal, C., El Homrani, M., Mingorance Estrada, Á.C.: Impact on the virtual learning environment due to COVID-19. *Sustainability* **13**(2), 582 (2021)
25. Ujkani, B., Minkovska, D., Nakov, O.: Understanding student success prediction using SHapley Additive exPlanations. In: 2023 International Scientific Conference on Computer Science, pp. 1–4. IEEE (2023)
26. Wang, C., Chang, L., Liu, T.: Predicting student performance in online learning using a highly efficient gradient boosting decision tree. In: Shi, Z., Zucker, J. (eds.) IIP 2022, LNCS, vol. 643, pp. 508–521. Springer, Cham (2022)
27. Waheed, H., Hassan, S.-U., Nawaz, R., Aljohani, N.R., Chen, G., Gasevic, D.: Early prediction of learners at risk in self-paced education: a neural network approach. *Expert Syst. Appl.* **213**, 118868 (2023). <https://doi.org/10.1016/j.eswa.2022.118868>